A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1 General physiological solvent:** most biological reactions take place between substances dissolved in water **1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes **1** High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) **1 Anomalous expansion:** ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) **1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3 **1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water **1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels **1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** All greenhouse gases expressed as CO2 equivalent (allowing for different warming strengths) **1** Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) **1** A fair share of the infrastructure: roads, car parks, signals, street lighting **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels **1** Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
02.3	**1 mark each** 1 mark for each of the 5 points below. **1** Hardy, disease-resistant Boran cattle – regular tick dipping reduces ticks for wild herbivores too **1** Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth **1** Electric fences keep game in and poachers out – biological corridors let animals (except rhinos) move to other reserves – problem elephants relocated **1 Monitoring:** radio-collared lions, drones, camera traps, track surveys – endangered species (Grevy's zebra, northern white rhino) protected **1 Community links:** ecotourism income funds health, education and water projects – local employment – cattle breeding help for local farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Overall decline from 62% to 22% **1** Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** The polyps expel their symbiotic zooxanthellae so the coral loses colour – bleached corals are alive but grow very slowly and may die if algae do not return **1** Raised water temperature (climate change / El Niño) **1 Pollution:** sewage, pesticides, cyanide, sunscreen – low oxygen from zooplankton blooms after overfishing **1** High light / solar activity – ocean acidification hinders skeleton formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps **1** Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release **1** Physical damage from anchors, divers, souvenir collection – loss of mangroves / seagrass nurseries that trap sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
03.4	1 mark for the point below. **1** Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Other MEDCs (USA, Canada, Australia) also had surpluses – Japan already had suppliers **1** Eastern Europe / USSR needed food but could not pay **1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Quotas limiting production (eg milk) **1** Set-aside payments for taking land out of production **1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) **1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
04.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape **1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat **1** Beetle banks of tussocky grass - natural pest predators **1** Skylark nesting plots – wild bird seed / nectar plants - winter food and pollinators **1** In-field tree protection – controlled stocking density to reduce compaction – ditch management – dry-stone walls – woodland management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds **1** Water-repelling surface of lotus flowers **1** Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to **1** Hooked burrs on seeds that stick to animal fur **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** The mound absorbs sunlight and warms – hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) **1** Air blowing over the mound has a lower pressure which also draws air out **1** Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Marsupials:** young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems **1** Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns **1** Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.4	1 mark for the point below. **1 They are light but strong:** hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
07.1	**1 mark each** 1 mark for each of the 3 points below. **1 Hard release:** individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) **1 Soft release:** gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds **1 Problems faced:** finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
07.2	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Large enough suitable habitat for a viable population **1** Reliable food supply **1** Low predation risk / original threat removed **1** Suitable breeding sites **1** Water **1** Support of the local human population **1** Official / legal protection of habitat and species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	5

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed **1** Identifies causes of failure so future releases can be improved / extra support given **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
08.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2** Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met **up to 2 Single species:** easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall **up to 2** Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them **up to 2 Simple age structure:** uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers **up to 2 Mixed woodland:** varied niches, stratification, dead wood, indigenous trees each with their own dependent species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	6
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline **1** Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp **1** Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Extensive cereal 5.0 – intensive cereal 3.0 **1** Intensive beef 0.1 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	2
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) **1** Each extra unit of input gives a smaller extra yield (law of diminishing returns) **1** Beef has very low productivity and efficiency because energy is lost at an extra trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
09.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Any input that increases productivity but requires the use of energy (usually fossil fuel) **up to 2** Manufacture of nitrate fertiliser (Haber process) and pesticides – fuel for machinery – pumping irrigation water – grain drying – food processing and transport – embodied energy of machinery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases **1** Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	

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